

Reader chips

give me the datasheets and comparison for the following RFID reader chips. e710, e518, e310, r2000, r500

I'll compare them as UHF RFID reader ICs/modules and I'll verify the exact part names first, because "E518" may be a vendor shorthand or typo for Impinj E510/E710 family rather than a standard chip name.

Thought for 16s

Below is a practical datasheet-style comparison for the RFID reader chips you listed. One note first: I could not verify a standard **"E518"** Impinj reader chip. I think you may mean **E510**. Impinj's current E-family reader chips are **E910, E710, E510, and E310**, and the public comparison tables list **E710 / E510 / E310 / R2000 / R500** together.

Datasheet / product document links

Chip	Datasheet / document
Impinj E710	E910/E710/E510/E310 reader chip datasheet:
Impinj E510	E510 product brief: and full E-family datasheet:
Impinj E310	Full E-family datasheet:
Impinj R2000	R2000 product brief:
Impinj R500	R500 product brief:
Portfolio comparison table	Impinj reader chip portfolio spec table:

Main comparison table

Item	E710	E510 / likely "E518"	E310	R2000	R500

Generation	Newer E-series	Newer E-series	Newer E-series	Older Indy series	Older Indy series
Air protocol	RAIN RFID / ISO 18000-63 / EPCglobal Gen2v2	Same	Same	Same	Same
Receive sensitivity	-88 dBm	-82 dBm	-75 dBm	-84 dBm	-68 dBm
Max read rate	~950–1000 tags/s	~550–700 tags/s	~250 tags/s	~900 tags/s	~190 tags/s
Typical power consumption	~0.5 W	~0.5 W	~0.5 W	~1.5 W	~1.1 W
Package	56-pin QFN	56-pin QFN	56-pin QFN	64-pin QFN	64-pin QFN
Package size	6 × 6 mm	6 × 6 mm	6 × 6 mm	9 × 9 mm	9 × 9 mm
Self-jammer cancellation	Yes	Yes	Yes	Yes	No / not listed in portfolio table
Reader modes	11	9	5	4	4
Integration level	Radio + modem + MAC + baluns + power detectors	Same	Same	Radio + modem	Radio + modem

Pin/software compatibility	Compatible with E710/E510/E310 family	Same	Same	Compatible with R2000/R500 family	Same
New design status	Recommended	Recommended	Recommended	Not recommended for new designs	Not recommended for new designs

The strongest simple takeaway: **E710 is the best overall, E510 is the balanced middle choice, E310 is the low-cost/shorter-range choice**, and **R2000/R500 are older parts that Impinj marks as not recommended for new designs**.

User-friendly explanation

1. Impinj E710

The **E710** is the best choice when you need long range, high sensitivity, and high tag population performance. It is suitable for fixed readers, handheld readers, shelves, cabinets, conveyors, gates, and industrial inventory systems.

Use **E710** when you want:

Need	Good fit?
Long read range	Excellent
Many tags at once	Excellent
High read speed	Excellent
New product design	Excellent
Low power compared with R2000	Excellent

Best for: warehouse portals, smart shelves, conveyor readers, cabinet readers, logistics, high-performance handhelds.

2. Impinj E510 / likely “E518”

I could not confirm a real standard **E518** reader chip, so I am treating this as **E510** unless you confirm otherwise.

The **E510** is the middle-performance E-series chip. It is designed for mobile readers, wearable readers, POS devices, smart appliances, and embedded IoT readers. It gives better performance than E310 but is cheaper/lower power than high-end designs using E710.

Use **E510** when you want:

Need	Good fit?
Medium read range	Very good
Battery-powered device	Very good
POS / mobile reader	Very good
Lower cost than E710	Good
High tag population	Good, but not as strong as E710

Best for: handheld readers, POS readers, wearable RFID devices, compact IoT readers, smart cabinets with medium range.

3. Impinj E310

The **E310** is the lower-cost E-series option. It is still a modern chip, but it is designed more for close-range or smaller tag population applications.

Use **E310** when you want:

Need	Good fit?
Low cost	Excellent
Short/medium read range	Good
Small group of tags	Good
High tag population	Not ideal
New design	Good

Best for: printers, kiosks, access control, desktop readers, small embedded readers, close-range devices.

4. Impinj R2000

The **R2000** was a very popular high-performance older chip. Many UHF RFID modules still use it. It has good sensitivity and high tag read rate, but it consumes more power and uses an older, larger package than the E-series. Impinj's comparison table says **R2000 is not recommended for new designs**.

Use **R2000** when:

Need	Good fit?
Existing module design	Good
Legacy product support	Good
New product design	Not recommended
High performance but older platform	Good
Low power	Not ideal

Best for: existing commercial modules, replacement designs, legacy readers, older fixed readers.

5. Impinj R500

The **R500** is the older low-cost Indy-series chip. It is mainly for short-range/proximity RFID readers. It has much lower sensitivity and lower read rate than E710/E510/R2000.

Use **R500** when:

Need	Good fit?
Very low-cost legacy design	Good
Short-range reader	Good
Many tags / long range	Poor

New product design	Not recommended
Low performance acceptable	Good

Best for: older POS readers, proximity readers, smart mirrors, access control, short-range embedded readers.

Recommendation

For a **new RFID reader product**, I would choose:

Application	Recommended chip
Long-range fixed reader / warehouse gate / conveyor	E710
Handheld reader / battery-powered reader / POS	E510
Low-cost close-range embedded reader	E310
Existing R2000 module replacement only	R2000
Existing low-cost legacy reader only	R500

My recommendation: **do not start a new design with R2000 or R500** unless you already have a module, PCB, firmware, or supplier locked to that platform. For new development, use **E710 / E510 / E310**.

okay, give me the download link for the datasheet pdf

Thought for 4s

Here are the **direct datasheet PDF download links**.

Chip	PDF download link
E710	Impinj E910 / E710 / E510 / E310 RAIN RFID Reader Chip Datasheet PDF

E518	I could not find an official E518 chip. It is likely E510 . Use this: Impinj E510 RAIN RFID Datasheet PDF
E310	Impinj E910 / E710 / E510 / E310 RAIN RFID Reader Chip Datasheet PDF
R2000	Impinj Indy R2000 Product Brief PDF
R500	Impinj Indy R500 Product Brief PDF
All-chip comparison	Impinj Reader Chip Portfolio Comparison PDF

Also, Impinj's support page confirms the latest public datasheet is for **E910, E710, E510, and E310** reader chips, not E518.

give me the recent price for each chip

Thought for 31s

Yes — but important point: **bare Impinj reader chip prices are not widely published**, especially for **E710 / E510 / E310**. Most public prices are for **reader modules or finished readers using those chips**, not the IC chip alone.

Recent public price estimate

Chip	Recent public price / estimate	Price confidence	Notes
Impinj E710	No reliable public bare-chip price found. E710-based modules/readers are around \$80–\$235+	Medium for module price, low for chip price	RFIDTagWorld lists an E710 UHF reader module sample price at \$80 ; GoToTags lists an E710 USB reader at \$234.64 , or \$214.73 at quantity 50.

Impinj E518	Not found	Low	I still think this is probably E510 , because Impinj's public E-series is E910 / E710 / E510 / E310 , not E518.
Impinj E510	No reliable public bare-chip price found. E510-based modules/readers are around \$125–\$335	Medium for module price, low for chip price	RFIDTagWorld lists an E510 integrated reader at \$125 ; Alibaba product search shows E510 reader/module listings around \$335 , but marketplace listings should be verified carefully.
Impinj E310	No reliable public bare-chip price found. E310-based readers are around \$115–\$200	Medium for module/reader price, low for chip price	GoToTags lists an E310 desktop USB reader at \$199.93 , or \$194.97 at quantity 10; RFIDTagWorld lists an E310 integrated reader at \$115 .
Impinj R2000 / IPJ-R2000	Around \$78–\$176 per chip in public broker/distributor listings	Medium-low	FindIC showed 100 pcs at \$7,844.30 , about \$78.44 each before tax ; DigiPart listings show around \$175 each from broker-type suppliers.
Impinj R500 / IPJ-R500	Around \$22.79–\$58.97 per chip in public listings	Medium	ElectronicsDatasheets shows Utsource at \$22.79 and Avnet Europe around \$58.97 ; OMO-IC shows bulk pricing

			around €16.99 at 1000 pcs , but stock/availability may vary.
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Simple buying expectation

Chip	Practical budget expectation
E710	Ask distributor; module/reader public price commonly \$80–\$235+
E510	Ask distributor; module/reader public price commonly \$125–\$335
E310	Ask distributor; module/reader public price commonly \$115–\$200
R2000	Bare chip public broker price roughly \$78–\$176
R500	Bare chip public price roughly \$23–\$59

For a real BOM calculation, I would treat the **E-series bare chip prices as RFQ-only** and use these rough relative levels:

E710 = highest price

E510 = middle price

E310 = lowest price

R2000 = old but still expensive in broker market

R500 = cheaper legacy chip